

Report: iNaturalist

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1. Introduction to Dataset

The dataset that I used for this project is from iNaturalist, a massive platform for nature lovers to record animals and publish their observations online. In this particular dataset there is a focus on the Barn Swallow Species and the dataset contains compilation of observations and their associated information from iNaturalist. The dataset is composed of 31281 entries (rows) and 63 columns on which include various types of information including the date of observation, time of observation, information on user, the taxonomy of the bird species, media information etc. See below for a full list of columns. The data from iNaturalist is accessible with the export module and allows all users to download biodiversity data. Users also make large contributions to new data with new records of sights of various species through human eyes. For this reason, iNaturalist is now one of the largest biodiversity databases.

Table 1: column list with descriptions from RAW data

Column Name	Description
observation	Unique identifier for each observation entry.
observed_on_string	The observation date in string format, including time if provided, no consistency - later removed
observed_on	Parsed date of the observation in YYYY-MM-DD format.
time_observed_at	The specific time of the observation in UTC.
time_zone	Time zone of the observation. Later removed as location is indicated
user_id	Unique identifier of the user who recorded the observation.
user_login	Username of the observer on iNaturalist.
created_at, updated_at	Timestamp when the observation was created and last updated on iNaturalist
quality_grade	Indicates the quality level of the observation, e.g., research, casual, etc.
license	The licensing applied to the observation data.
url	Website Link to the observation on iNaturalist.
image_url, sound_url	Links to associated images and sound recordings for the observation.
tag_list	Tags associated with the observation.

description	Free-text description or notes provided by the observer.
num_identification_agreements, num_identification_disagreements	Count of agreements and disagreements about the species identification.
captive_cultivated	Boolean indicating whether the organism is in a controlled environment.
place_guess	Location name as guessed or specified by the user.
latitude, longitude	Coordinates of the observation location.
positional_accuracy	Accuracy of the location data in meters.
geoprivacy, taxon_geoprivacy	Privacy settings for geolocation data.
coordinates_obscured	Boolean indicating whether coordinates have been obscured for privacy.
positioning_method, positioning_device	Information about how the location was determined (e.g., GPS).
place_town_name, place_county_name, place_state_name, place_country_name	Detailed location breakdown by administrative divisions.
species_guess	Initial species guessed by the observer.
scientific_name	Verified scientific name of the species.
common_name	Common name of the species.
iconic_taxon_name	Higher-level taxonomic category (e.g., Aves).
taxon_id	Unique identifier for the taxonomic entry.
Taxonomic Hierarchy Columns	Detailed classification fields: kingdom, phylum, class, order, family, genus, species, subspecies, etc.

2. Ontology search

Before starting with the data transformation process, I explored the iNaturalist website to gain a better understanding of how its database operates and the type of information it stores. This allowed me to understand the meanings of the dataset columns and their context within biodiversity data. Then, to ensure the data transformation aligns with established semantic standards, I researched existing ontologies specializing in taxonomy

and biodiversity data and for general ontology as well. These include: Darwin core (dwc) : <http://rs.tdwg.org/dwc/>, Dublin Core (dc) : <http://purl.org/dc/elements/1.1/>, schema.org (sc) : <https://schema.org/> (in my UML model and Protege, I use 'sch' as prefix instead of sc as in Protege it was unable to identify 'sc' as schema.org despite giving its prefix definition and led to a different url), Dublin Core Terms (dcterms): <http://purl.org/dc/terms/> and lastly GeoNames (Geo): URL: <http://www.geonames.org/ontology>. DarwinCore has its domain on biodiversity information and contains vocabulary for describing occurrences of certain species, description taxonomy, and their associated metadata regarding location and date etc. such as : dwc:eventDate, dwc:scientificName. I also used schema.org and Dublin Core dictionary for general-purpose vocabulary and for metadata to describe 'Person' , 'Place', 'Taxon', 'Media' , 'Location' etc. Dublin Core Terms is an extension of Dublin Core Terms and contains more specific terms such as dcterms:license. Then lastly, GeoNames was used to get ontologies on geographical attributes such as latitude and longitude.

3. Data Transformation

The data transformation process was done in accordance with the selected ontology terms and aimed to refine the dataset for consistency, clarity, and semantic alignment. First, columns that contained overly specific, with most rows containing null values, were removed. These columns were mostly about highly specific taxonomy levels such as 'subtribe' columns which were recorded only in a few entries. Some other columns that were omitted are 'geoprivacy' and 'positioning device' which were also specific information and were less relevant to other data. Some columns were aligned or were renamed to better align with corresponding ontology terms, enhancing their semantic meaning.

Data formatting was also done on date attributes including: 'observed_on', 'created_at', 'updated_at' columns which were standardized to the YYYY-MM-DD format from a string format. For the 'time observed at' column, times were extracted where available and were reformatted to 00:00:00 from YYYY-MM-DD 00:00:00 format. Fields with HTML tags, such as descriptions, were cleaned to retain only plain text, to ensure readability. Some data columns were found to contain overlapping or identical values, resulting in redundancy, and were therefore omitted to streamline the dataset. For example, certain date-related columns provided the same information in different formats—one as a string and another in a structured date format. In such cases, the more structured format was retained, and the string representation was discarded to eliminate duplication.

4. UML Schema

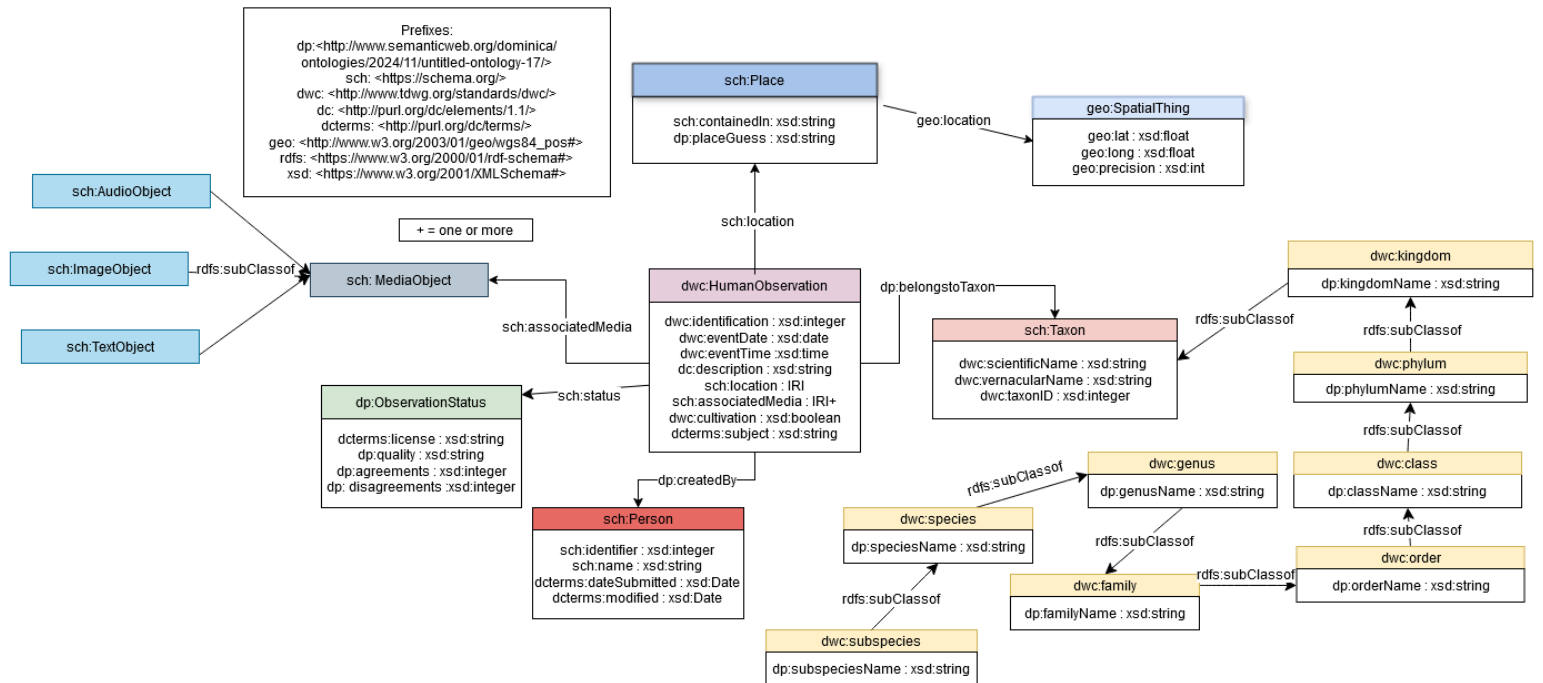


Figure 1. UML diagram modelling the semantic relationships and data properties of classes within the ontology, including classes: **dwc:HumanObservation**, **sch:Taxon**, **sch:Place**, **geo:Location**, **sch:MediaObject**, and **sch:Person** and their attributes and subclasses.

Below, there is an explanation of all my classes and their attributes:

Class: dwc:HumanObservation

Label: Human Observation

Explanation: This class represents observations made by humans and contains details of observation identification, date, time, description, associated location, media, whether the animal has been cultivated or not (dwc:cultivation) and tags used in the observation (dcterms:subject).

Attributes (DataProperties except for bold attributes):

- dwc:identification : xsd:integer (Unique identifier for the observation)
- dwc:eventDate : xsd:date (date when the observation was created)
- dwc:eventTime : xsd:time (time when the observation was made)
- dc:description : xsd:string (Description of the observation)
- **sch:location : IRI** (IRI pointing to a location instance - sch:Place)
- **sch:associatedMedia : IRI+** (IRIs linking to multiple associated media objects - sch:MediaObject)
- dwc:cultivation : xsd:boolean (status of cultivation of an animal, true if cultivated, false if found in wild)
- dcterms:subject : xsd:string (subject tags used in the observation record)

Class: sch:Taxon

label:Taxon

Explanation: This class represents the hierarchical classification of biological organisms that contain taxonomic details including scientific and vernacular(common) names, taxonomy identifiers, and ranks. This class contains 8 hierarchical subclasses to represent different levels of taxonomy.

Attributes (DataProperties except for bold attributes):

- dwc:scientificName : xsd:string (Scientific name of the taxon)
- dwc:vernacularName : xsd:string (Common name of the taxon)
- dwc:taxonID : xsd:string (Identifier for the taxon)

Subclasses and Attributes:

1. Subclass of Taxon: dwc:kingdom
Label: Kingdom
 - dp:kingdomName : xsd:string
(The name of the kingdom to which the taxon belongs.)
2. Subclass of Kingdom: dwc:phylum
Label: Phylum
 - dp:phylumName : xsd:string
(The name of the phylum to which the taxon belongs.)
3. Subclass of Phylum: dwc:class
Label: Class
 - dp:className : xsd:string
(The name of the class to which the taxon belongs.)
4. Subclass of Class: dwc:order
Label: Order
 - dp:orderName : xsd:string
(The name of the order to which the taxon belongs.)
5. Subclass of Order: dwc:family
Label: Family
 - dp:familyName : xsd:string
(The name of the family to which the taxon belongs.)
6. Subclass of Family: dwc:genus
Label: Genus
 - dp:genusName : xsd:string
(The name of the genus to which the taxon belongs.)
7. Subclass of Genus: dwc:species
Label: Species
 - dp:speciesName : xsd:string
(The name of the species to which the taxon belongs.)
8. Subclass of Species: dwc:subspecies
Label: Subspecies
 - dp:subspeciesName : xsd:string
(The name of the subspecies to which the taxon belongs.)

Class: sch:Place**Label:Place**

Explanation: Contains information of spatial entities such as for buildings, locations and geographical areas.

Attributes (DataProperties except for bold attributes):

- sc:containedInPlace : xsd:string (has a hierarchical relationship between different places. For example a city(place) is found in a country(place))
- dp:placeGuess : xsd:string (an approximation of the place name)

Class: geo:SpatialThing**Label:SpatialThing**

Explanation: describes more specific details on the location through geographical points.

Attributes (DataProperties except for bold attributes):

- geo:lat : xsd:float (Latitude of the location)
- geo:long : xsd:float (Longitude of the location)
- geo:precision : xsd:int (precision of the location)

Class: sch:MediaObject**label:MediaObject**

Explanation: Represents any form of media associated with an observation, such as audio, images, or text.

Subclasses of sch:MediaObject**1. Subclass: sch:AudioObject**

Label: Audio Object

Represents audio media associated with an observation, such as recorded sounds.

2. Subclass: sch:ImageObject

Label: Image Object

Represents image media associated with an observation, such as photos or illustrations.

3. Subclass: sch:TextObject

Label: Text Object

Represents textual media associated with an observation

Class: sch: Person**label:Person**

Explanation: The Person class represents an individual who contributes to or interacts with the observation data. This could be a user who uploads, modifies, or is otherwise associated with the data. The attributes capture essential details about the person and their interactions with the observations.

Attributes (DataProperties except for bold attributes):

- sch:identifier : xsd:string (Unique identifier for the user)
- sch:name : xsd:string(IRI pointing to observations created by the user)
- dcterms:dateSubmitted : xsd:Date (the date of when the person uploaded their observation)
- dcterms:modified :xsd:Date (the date of when the person modified and updated their observation)

Class: dp:ObservationStatus**Label: Observation Status**

Explanation: The ObservationStatus class provides metadata about the quality and consensus regarding an observation. It includes attributes for licensing, academic utility, and community feedback. This class helps track the reliability and relevance of observations

Attributes (DataProperties except for bold attributes):

- dcterms:license :xsd:string (License information regarding the observation and the taxonomy of the organism)
- dp:quality :xsd:string (the level of academic usage from the observation - example: research)
- dp:agreements:xsd:integer (number of agreements by other users)
- dp:disagreements :xsd:integer (number of disagreements by other users)

Relationships (Main Object Properties - sub object properties not mentioned in this report)

- 1. Object Property: sch:associatedMedia**
Label: Associated Media
Domain: dwc:HumanObservation
Range: sch:MediaObject
Explanation: Links an observation to one or more media objects (e.g., images, audio, or text) that are associated with it.
- 2. Object Property: sch:status**
Label: Status
Domain: dwc:HumanObservation
Range: dp:ObservationStatus
Explanation: Indicates the status of the observation, including its quality, agreements, and disagreements.
- 3. Object Property: dp:belongstoTaxon**
Label: Belongs to Taxon
Domain: dwc:HumanObservation
Range: sch:Taxon
Explanation: Associates an observation with a specific taxon, identifying the biological classification of the entity observed.
- 4. Object Property: geo:location**
Label: Location
Domain: sch:Place
Range: geo:SpatialThing
Explanation: Specifies the geographic coordinates (latitude and longitude) of a place.
- 5. Object Property: sch:location**
Label: location
Domain: dwc:HumanObservation

Range: sch:Place

Explanation: Links an observation to the place where it was recorded or observed.

6. Object Property: dp:createdBy

Label: Created By

Domain: dwc:HumanObservation

Range: sch:Person

Explanation: Indicates the person responsible for creating or uploading the observation.

7. Object Property : sch:containedIn

Label: Contained In

Domain: sch:Place

Range :sch:Place

Explanation: Indicates the place that belong to place

5. SPARQL Codes

The following prefixes are used throughout the Sparql:

PREFIX ns1: <http://example.com/base/>

PREFIX ns11: <http://example/base/>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

PREFIX geo: <http://www.w3.org/2003/01/geo/wgs84_pos#>

PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

PREFIX schema: <http://schema.org/>

PREFIX dwc: <http://dwc.tdwg.org/terms/>

PREFIX dc: <http://purl.org/dc/elements/1.1/>

PREFIX dcterms: <http://purl.org/dc/terms/>

PREFIX dp:

<http://www.semanticweb.org/dominica/ontologies/2024/11/untitled-ontology-17/>

1. This query lists all observations with their corresponding location and coordinates.

```
SELECT ?observation
```

```
    ?locationName
```

```
    ?latitude
```

```
    ?longitude
```

```
WHERE {
```

```
  # Observation details
```

```
  ?observation a dwc:HumanObservation ;
```

```
    schema:location ?location .
```

```
  # Location details
```

```
  ?location dp:placeGuess ?locationName ;
```

```
geo:location ?spatialThing .
```

```
# Spatial details
```

```
?spatialThing geo:lat ?latitude ;  
                geo:long ?longitude .
```

```
}
```

Result

observation	locationName	latitude	longitude
ns1:observation296	Nisqually Washington	47.0567016602	-122.702003479
ns1:observation525	Point Reyes National Seashore, Marin Co., CA	37.9702987671	-122.7910003662
ns1:observation224 5	King City, MO	40.053691864	-94.5237503052

2. Number of observations made in a place for a specific species

This query retrieves observations grouped by their location, counts the number of observations for each location, and orders the results by the observation count in descending order.

```
SELECT ?locationName (COUNT(?observation) AS ?observationCount)  
      (GROUP_CONCAT(DISTINCT ?speciesName; separator=", ") AS  
?speciesObserved)
```

```
WHERE {
```

```
# Observation details
```

```
?observation a dwc:HumanObservation ;  
              schema:location ?location ;  
              dp:belongstoTaxon ?taxon ;  
              dwc:eventDate ?eventDate .
```

```
# Location details
```

```
?location dp:placeGuess ?locationName .
```

```
# Taxon details
```

```
?taxon dwc:scientificName ?speciesName .
```

```
# Filter to include only observations submitted after 2015
```

```
FILTER (xsd:date(?eventDate) > "2015-01-01"^^xsd:date)
```

```
}
```

GROUP BY ?locationName
ORDER BY DESC(?observationCount)
LIMIT 3

Result

locationName	observationCount	speciesObserved
United Kingdom, England, Morpeth	318	Hirundo rustica
Mexico, Coahuila, Saltillo	288	Hirundo rustica, Passer domesticus
France, Gironde, Anglade	286	Hirundo rustica

3. General retrieval of selected data

This SPARQL query retrieves details about bird observations, specifically including the species observed, the location of the observation, the date the observation was made, the observer (person), and the type of event associated with the observation. The query filters the data to include only observations that occurred after January 1, 2015 for relevancy to newer data.

```
SELECT ?speciesName
      ?locationName
      ?eventDate
      ?observerName
      ?qualityType
WHERE {
  # Observation details
  ?observation a dwc:HumanObservation ;
    schema:location ?location ;
    dp:belongstoTaxon ?taxon ;
    dwc:eventDate ?eventDate ;
    schema:name ?observerName ;
    dp:quality ?qualityType .

  # Location details
  ?location dp:placeGuess ?locationName .

  # Taxon details
  ?taxon dwc:scientificName ?speciesName .

  # Filter to include only observations submitted after 2015
  FILTER (xsd:date(?eventDate) > "2015-01-01"^^xsd:date)
}
LIMIT 3
```

Result:

speciesName	locationName	eventDate	observerName	qualityType
Hirundo rustica	Stanley Park, Liverpool, England, GB	2019-07-06	cazathome	research
Sturnus vulgaris	Sorøvej 172, Fjenneslev, Sjælland, DK	2019-07-06	martinmalmjohansen	research
Hirundo rustica	Knowsley, UK	2019-07-06	croxvols	Research

4. Count the number of media

```

SELECT ?observation
      ?speciesName
      (COUNT(?media) AS ?numMedia)
WHERE {
  # Observation details
  ?observation a dwc:HumanObservation ;
    dp:belongstoTaxon ?taxon ;
    schema:associatedMedia ?media .

  # Taxon details
  ?taxon dwc:scientificName ?speciesName .
}
GROUP BY ?observation ?speciesName
HAVING (COUNT(?media) > 0)
LIMIT 3

```

Result

observation	speciesName	numMedia
ns1:observation27065277	Hirundo rustica	1
ns1:observation27066114	Hirundo rustica	1
ns:1observation27066277	Hirundo rustica	1

5. Number of agreements by observation

This query retrieves information about human observations and associate them to the observation's date, time, and the total number of agreements associated with the observation, and ranks them in descending order of total agreements

```

SELECT ?observation
      ?eventDate
      ?eventTime
      (SUM(?numberOfAgreements) AS ?totalAgreements)
WHERE {
  # Observation details
  ?observation a dwc:HumanObservation ;
    dwc:eventDate ?eventDate ;
    dwc:eventTime ?eventTime ;
    dwc:identification ?identification ;
    dp:belongstoTaxon ?taxon ;
    schema:location ?location ;
    schema:associatedMedia ?media ;
    dwc:numberOfAgreements ?numberOfAgreements .
}
GROUP BY ?observation ?eventDate ?eventTime
ORDER BY DESC(?totalAgreements)
LIMIT 3

```

Result:

observation	eventDate	eventTime	totalAgreements
ns1:observation56752	2009-06-06	13:16:00	9
ns1:observation52217	2010-05-07	N/A	9
ns1:observation52737	2011-06-29	N/A	9

6. Areas of Improvements:

These sample queries are useful in extracting important information from various biodiversity data. By querying species occurrence data, such as the location and time of sightings, we can gain valuable insights of geographical distribution of species which can then help identify different patterns, such as species that are expanding or contracting their ranges, or changes in migration patterns due to climate change. SPARQL queries can help assess the quality of data by identifying inconsistencies, missing information, or potential errors. For example, it can be used to filter out observations with incorrect species identification or many missing attributes or data values. This will help ensure that only reliable data is included in analyses. To extend on this, the data source can be connected to external resources to further enhance the queries. For example, **GBIF (Global Biodiversity Information Facility)** provides occurrence data of numerous species that can be used to complement iNaturalist observations or be used for comparisons. By integrating data from both platforms, researchers can get a more comprehensive view of species distributions globally. **GeoNames** can also be used further to gain a more detailed understanding of the species

habitat and attributes such as temperature, humidity etc. These detailed databases of geographical locations can be linked to iNaturalist observations to expand the knowledge on where species are found in relation to specific administrative regions, climate zones, or geographic features like mountains or rivers.

I haven't thought about this before but to integrate data, I can try to employ properties like owl:sameAs to link iNaturalist observations to external data as such: <observation296> owl:sameAs <<http://www.gbif.org/species/2492487>>. Moreover, federated queries can be used to query data from multiple data sources in a single query. I tried to make federated queries between iNaturalist and GBIF to create a more enhanced dataset, however came across some errors which I couldn't solve in time. If a federated query was made, the species observations from both iNaturalist and GBIF, would have been compared or merged.

Additional things that could be done is to create a visual representation of the data. In this specific context, a visual **map** could be used to visualise the different distributions of the species around the world. Tools like QGIS can be used to visualize these relationships geographically. For occurrences of the species, or for seeing contributing users, Bar Charts can be used to see the frequency of species occurrences or user contribution by time or location. This helps in identifying trends, such as which species are more commonly observed in particular seasons or regions.